More difficult parallel and perpendicular lines

What to do

1 Which pairs of lines are parallel?

a $y = \frac{1}{2}x - 3,\ y = \frac{1}{2}x + 4$

b $y = 3x - 8,\ y = \frac{-1}{3}x + 7$

c $y = \frac{3}{10}x + 2,\ y = \frac{3}{10}x - 3$

d $2y - 3x = 6,\ 4y - 6x = 6$

2 Which pairs of lines are perpendicular?

a $y = \frac{1}{2}x - 3,\ y = -2x + 4$

b $y = 3x - 8,\ y = \frac{-1}{3}x + 7$

c $y = \frac{3}{10}x + 2,\ y = \frac{10}{3}x - 3$

d $2y - 3x = 6,\ 6y - 4x = 6$

3 Find the equation of the line that is parallel to $4x - 5y - 8 = 0$ and passes through the origin.

4 Find the equation of the line that is perpendicular to the line $y - 2x + 7 = 0$ and passes through the point (3, 5).

5 In the triangle joining the points O(0, 0), A(5, −1) and B(−6, −7), prove that AB is parallel to the line joining the midpoints of OA and OB.

6 If $ax - by + d = 0$ and $px - qy - n = 0$ are perpendicular, find a relationship between a, b, p and q.